

Technical Document #725: Natural Language Processing - Comprehensive Overview

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Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Coreference resolution identifies all expressions that refer to the same entity in a text. It is essential for understanding document-level coherence.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Text summarization generates a concise summary of a longer text.
- Question answering systems extract answers from a given context.
- Word embeddings represent words as dense vectors in continuous space.